Evaluating Machine Learning Algorithms for Gait Anomaly Detection Using Publicly Available Wearable Sensor Dataset

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I. Introduction

A. Research Context and Motivation

Human gait analysis is an essential area within clinical assessment, rehabilitation engineering, and assistive technology development. It provides non-invasive knowledge of neurological function, musculoskeletal health, and fall risk assessment. The adoption of wearable sensor technologies, particularly Force Sensitive Resistors (FSRs) and Inertial Measurement Units (IMUs), has significantly advanced the field by delivering continuous, real-time monitoring of gait characteristics outside clinical settings [1], [2]. Traditional gait analysis methods rely on expensive, laboratory-based motion capture systems that require specialised equipment and trained personnel, limiting accessibility and scalability. In contrast, wearable sensors embedded in insoles or attached to the body present a portable and cost-effective solution for collecting kinematic and kinetic data during natural walking [3], [4]. The ability to detect deviations from typical gait patterns has important implications for early diagnosis of conditions such as Parkinson's disease, monitoring stroke recovery, assessing fall risk in older adults, and tracking rehabilitation progress following surgery [5]. The importance of gait anomaly detection research has increased in recent years, driven by digitalisation in healthcare, demographic shifts toward elderly populations, and developments in machine learning. Wearable health monitoring devices have evolved from specialised research instruments to widely adopted consumer products, reflecting the growing demand for remote patient monitoring and preventive healthcare. Machine learning algorithms, especially supervised classification methods, have achieved notable success in analysing complex sensor data to detect subtle gait abnormalities that may not be apparent to human observers [1], [6]. Research indicates that integrated FSR-IMU sensor systems can achieve gait phase detection accuracies above 89% while supporting real-time processing suitable for embedded applications [7], [8]. The combination of accessible sensor technology, increased computational capabilities, and advanced algorithms offers significant opportunities for developing intelligent systems that improve clinical decision-making and facilitate proactive health interventions [9]. This study evaluates machine learning algorithms for detecting gait anomalies using publicly available datasets of wearable sensors. Instead of collecting primary data through custom hardware development, as explored in parallel dissertation work with an ESP32-based insole system using FSR and MPU6050 sensors, this research employs an alternative approach. It leverages validated datasets to enable rigorous evaluation and comparison of algorithms. This methodology eliminates hardware development overhead, focuses on computational methods and feature engineering, ensures reproducibility through standardised benchmark data, and allows systematic comparison of multiple machine learning techniques under controlled conditions [10]. The study specifically examines classification algorithms such as Support Vector Machines, Random Forests, and Neural Networks, assessing their performance in distinguishing normal gait patterns from various anomaly types using sensor data that include accelerometer, gyroscope, and pressure measurements.

B. Research Aim

This research systematically evaluates and compares the classification performance of Support Vector Machine, Random Forest, and Artificial Neural Network algorithms for detecting gait anomalies using publicly available wearable sensor datasets. The study aims to identify which algorithm achieves optimal accuracy, precision, recall, and F1-score under standardised experimental conditions.

C. Purpose Statement

This quantitative experimental study compares the classification performance of three machine learning algorithms: Support Vector Machine (SVM), Random Forest (RF) and Artificial Neural Network (ANN) to detect gait anomalies

using publicly available wearable sensor datasets. The independent variables are the type of machine learning algorithm (SVM, RF, ANN) and the feature extraction method (time-domain features, combined time-frequency features). The dependent variables are classification performance metrics, including accuracy, precision, recall, F1-score, and area under the ROC curve (AUC-ROC). The analysis will utilise publicly available wearable sensor datasets, such as the Multi-Modal Gait Database [10], which include normal and abnormal gait patterns with synchronised FSR and IMU sensor data, to determine which algorithmic approach provides optimal detection performance. The findings will guide software developers, biomedical engineers, and healthcare technology researchers in selecting algorithms based on evidence to implement gait-monitoring systems in clinical and assistive technology applications.

D. Research Question and Hypothesis

1) Research Question: Which machine learning algorithm (Support Vector Machine, Random Forest, or Artificial Neural Network) achieves the highest classification performance for detecting gait anomalies when trained on publicly available wearable sensor datasets?

2) Null Hypothesis (H_0): There is no statistically significant difference in classification performance (measured by F1-score) among Support Vector Machines, Random Forests, and Artificial Neural Networks when applied to gait anomaly detection using publicly available wearable sensor datasets.

3) Alternative Hypothesis (H_1): There is a statistically significant difference in classification performance (measured by F1-score) among Support Vector Machine, Random Forest, and Artificial Neural Network algorithms when applied to gait anomaly detection, with at least one algorithm demonstrating superior performance.

E. Research Positioning and Methodology Framework

This study's methodological design is structured according to Saunders et al.'s Research Onion framework [11], which systematically addresses philosophical assumptions, research approaches, strategies, and techniques from abstract to concrete levels. Figure 1 illustrates the research positioning across the layers of the Research Onion, with the selected methodological choices highlighted.

1) Research Philosophy: Positivism: This research adopts a **positivist epistemological stance**, assuming that gait patterns represent objective, measurable phenomena that can be quantified through sensor data and analysed using statistical methods. The positivist paradigm aligns with the natural sciences tradition, treating sensor measurements as reliable representations of physical reality and algorithm performance as objectively assessable through standardised metrics [12]. This philosophical position justifies the use of quantitative methods, hypothesis testing, and empirical validation through controlled experimentation. Studies utilising instrumented insoles and IMU sensors have consistently demonstrated the objective measurability of gait parameters across different populations and pathological conditions [5], [13], [14].

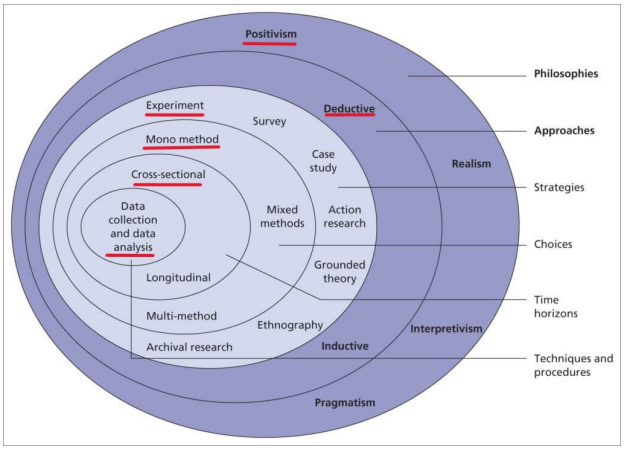

Fig. 1. Saunders' Research Onion showing the methodological positioning of this study. The highlighted layers indicate: Positivism (Philosophy), Deductive (Approach), Experimental (Strategy), Mono-method Quantitative (Methodological Choice), Cross-sectional (Time Horizon), and Secondary Data Analysis (Techniques and Procedures).

2) Research Approach: Deductive: This study adopts a deductive approach, commencing with established theories of machine learning classification and gait biomechanics, and subsequently formulating testable hypotheses regarding comparative algorithm results. The research advances from the general principle that machine learning can automate gait pattern recognition to specific predictions about which algorithmic methods will achieve better classification metrics on a standardised dataset [1], [6]. This deductive reasoning contrasts with inductive approaches, which derive theories from observed patterns in data. The deductive framework assists hypothesis testing using inferential statistics to determine whether the observed performance differences are statistically significant or due to random variation.

3) Research Strategy: Experimental: The research employs an **experimental strategy** involving systematic manipulation of independent variables (machine learning algorithm type, feature extraction methods) while measuring effects on dependent variables (classification performance metrics). This strategy enables causal inference about which algorithmic approaches are most effective for gait anomaly detection [1], [9]. The experimental design incorporates controlled conditions (all algorithms evaluated on identical datasets with consistent preprocessing), randomisation (cross-validation with random data partitioning), and replication (multiple experimental runs with different random seeds) to ensure robust and generalizable findings.

4) Methodological Choice: Mono-Method Quantitative: This research employs a **mono-method quantitative approach**, exclusively using numerical data analysis and statistical evaluation. The methodology is inherently quantitative because: (1) wearable sensor datasets consist of numerical time-series measurements (accelerometer readings in m/s^2, gyroscope readings in rad/s, pressure values in kPa) [4], [7], [8]; (2) machine learning algorithms operate on numerical

feature vectors and produce probabilistic classification outputs [1], [6]; (3) performance is assessed through quantitative metrics including accuracy, precision, recall, F1-score, and area under the ROC curve (AUC-ROC); and (4) statistical tests (e.g., paired t-tests, ANOVA, Friedman test) determine whether performance differences between algorithms are statistically significant. Qualitative methods are not employed, as the research questions concern measurable classification performance rather than subjective experiences or interpretive understanding.

5) Time Horizon: Cross-Sectional: This is a **cross-sectional study** analysing existing datasets at a single point in time, rather than longitudinally tracking changes over extended periods. The focus is on comparing algorithmic performance across different techniques at present, not examining the temporal evolution of gait patterns or algorithm capabilities.

6) Data Collection Techniques: Secondary Data Analysis: The study utilises **secondary data** from a publicly available wearable sensor dataset. This approach offers advantages, including access to validated, peer-reviewed dataset; reduced time and resource requirements compared to primary data collection; and enhanced reproducibility since other researchers can access the same data [10]. Recommended data sources include the Multi-Modal Gait Database [10], which provides synchronised data from 17 IMUs, 16 pressure sensors, and eye-tracking during natural urban walking, as well as other datasets from repositories such as Kaggle, UCI Machine Learning Repository, and PhysioNet. Data preprocessing will follow established protocols, including noise filtering, segmentation, normalisation, and feature extraction [4], [7]. Machine learning implementation will utilise Python-based libraries (scikit-learn, TensorFlow/Keras) following best practices for supervised classification.

7) Justification for Quantitative Secondary Data Approach: Using a publicly available dataset is methodologically sound and widely accepted in machine learning research for algorithm evaluation and benchmarking. This approach enables a rigorous comparison of algorithmic techniques without the confounding variables introduced by hardware variations, sensor calibration differences, or data-collection inconsistencies that arise from primary data collection. Furthermore, it allows this research to complement the dissertation's hardware-focused work by concentrating specifically on computational methods and algorithm optimisation [10].

II. Literature Review

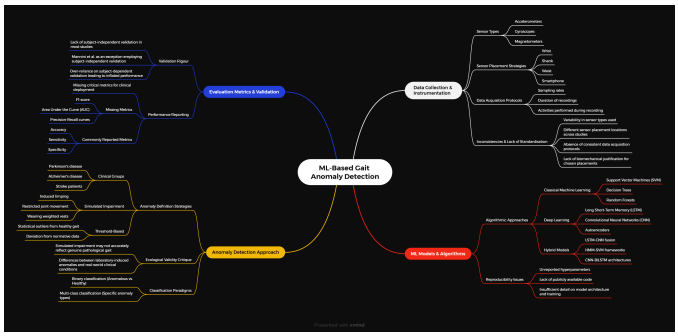

Fig. 2. Hierarchical Literature Map: Gait Anomaly Detection Using Wearable Sensors

Instrumentation and Data Collection Paradigms

Sensor placement strategies exhibit striking inconsistency across studies, undermining generalisability and cross-study comparison. Gao et al. [15] employ leg-mounted IMUs for Euler angle extraction, Mannini et al. [16] position sensors at shank and waist, whilst Kiprijanovska et al. [17] controversially adopt bilateral wrist-worn devices. This wrist-based approach, whilst convenient, fundamentally compromises biomechanical validity; wrist kinematics provide only distal, attenuated representations of lower-limb gait pathology, introducing substantial signal degradation and phase lag. Critically, none of these studies justify their placement decisions through biomechanical reasoning or comparative analysis, suggesting convenience-driven rather than evidence-based instrumentation. Furthermore, sampling frequencies remain unreported in four of five studies [15], [17], [18], precluding assessment of Nyquist compliance and aliasing risk—a methodological opacity unacceptable in signal processing research.

Dataset characteristics reveal more troubling limitations. Kiprijanovska et al. [17] simulate abnormalities using impairment glasses with merely 18 participants, raising profound ecological validity concerns; simulated impairment cannot replicate the neuromuscular complexity, compensatory strategies, or inter-individual variability inherent in genuine pathological gait. Sejdic et al. restrict data collection to treadmill walking, eliminating the turning, acceleration, and terrain variation essential to real-world gait assessment. Hemmatpour et al. [18] acknowledge creating a "realistic parameters" dataset but provide no participant demographics, clinical validation, or ground-truth labelling protocol, rendering their 93.5% accuracy claim methodologically unverifiable. Only Mannini et al. [16] employ clinically diagnosed populations (post-stroke, Huntington's disease), yet participant counts remain undisclosed, preventing power analysis or effect size estimation.

Algorithmic Approaches and Feature Engineering

The reviewed studies span a methodological spectrum from hand-crafted feature engineering to end-to-end deep learning, yet none adequately justify their algorithmic choices

against task requirements. Gao et al. [15] propose LSTM-CNN fusion (LCWSnet) for temporal-spatial feature learning but omit architectural specifics—layer depths, filter dimensions, dropout rates, and convergence criteria remain unreported, precluding reproducibility. Mannini et al. [16] combine Hidden Markov Models with Support Vector Machines, leveraging probabilistic sequence modelling, yet this hybrid approach introduces computational complexity and hyperparameter interdependencies without demonstrating superiority over simpler baselines. Kiprijanovska et al. [17] adopt CNN-BiLSTM for spatiotemporal learning from raw signals, claiming end-to-end feature extraction, but their 88.9% accuracy on binary classification of simulated data suggests potential overfitting to artificial patterns rather than genuine pathology detection.

Sejdic et al. present exhaustive time-frequency feature extraction across tri-axial accelerometry, yet their statistical analysis framework lacks classifier implementation, leaving feature utility unquantified beyond group differentiation claims. Hemmatpour et al. [18] advocate nonlinear predictive threshold models for computational efficiency, achieving 3.5× speed improvements, but threshold-based approaches inherently assume unimodal anomaly distributions and fixed decision boundaries—assumptions violated by the heterogeneous, context-dependent nature of gait pathologies. Critically, no study employs multi-modal sensor fusion (accelerometry, gyroscopy, magnetometry, pressure) despite evidence that sensor complementarity enhances robustness to noise and movement artefacts.

Evaluation Metrics and Validation Rigour

Performance reporting demonstrates alarming inconsistency and methodological weakness. Gao et al. [15] report "improved accuracy" without numeric values; Hemmatpour et al. [18] claim 93.5% accuracy without specifying class balance, baseline comparisons, or confidence intervals. Only Kiprijanovska et al. [17] report sensitivity (90.6%) and specificity (86.2%), yet their binary classification on simulated data provides limited clinical insight. Mannini et al. [16] achieve 90.5% subject-level classification via leave-one-subject-out cross-validation—the sole study employing subject-independent validation—yet majority voting aggregation may mask within-subject variability and temporal instability.

Crucially, no study reports F1-scores, area under ROC curves, or precision-recall analysis, metrics essential for imbalanced clinical datasets where false negatives carry severe consequences. Validation protocols remain opaque; cross-validation strategies, train-test splits, and temporal holdout procedures are largely unreported [15], [17], [18], raising overfitting concerns. Furthermore, none address model interpretability, feature importance, or failure mode analysis—critical omissions for clinical translation where algorithmic transparency underpins trust and regulatory approval.

Knowledge Gaps and Justification for Multi-Modal Quantitative Approaches

The reviewed literature exposes three critical gaps: (1) absence of standardised, multi-modal, clinically validated benchmark datasets; (2) lack of rigorous, subject-independent validation with comprehensive performance metrics; (3) insufficient biomechanical grounding in sensor placement and feature engineering. The Multi-Modal Gait Database (Losing & Hasenjäger, 2022) directly addresses these deficiencies by providing synchronised, multi-sensor data from validated clinical populations with standardised protocols, enabling reproducible, generalisable model development. Quantitative machine learning approaches, when coupled with subject-independent cross-validation and multi-metric evaluation, offer the methodological rigour absent in current literature, advancing gait anomaly detection from exploratory studies toward clinically deployable diagnostic tools.

III. Research Methodology

IV. Findings & Discussion of Results

V. Conclusion

Appendix A
Supporting material

You can add screen shots and statistics. Stick to essential information.

Acknowledgement

You can dedicate this section for special assistance that you were given by a 3rd party. Not your mentor, relatives or questionnaire participants.

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